



# The Chai Logs

A Case Study in Customer Behaviour and Discrete Distributions

STA201 – Theory of Probability and Statistics

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## The Script: A Story of Nikhil and Kavya

**Date:** 15 March 2027, 9:30 AM

**Location:** Chai Theory Headquarters, Mohali City Center, Mohali.

**Context:** Chai Theory is a direct-to-consumer (D2C) tea brand that sells curated single-origin teas through a subscription model. The company has about 5,000 active subscribers. Over the past quarter, the retention rate has dropped from 78% to 71%.

**The Team:**

- **Nikhil** — B.Tech CSE (3rd Sem), data analytics intern
- **Ms. Menon** — Head of Growth
- **Kavya** — a loyal subscriber whose behaviour is being analysed

**Ms. Menon:** Nikhil, our retention is falling. We send personalised messages to every subscriber, but response rates are dropping. I need to know when customers are most likely to respond.

**Nikhil:** I'll analyse the message logs.

**Ms. Menon:** Start with Kavya. She has been with us for 8 months, but her engagement is falling. Look at her last 60 days of data.

**Nikhil:** What should I look for?

**Ms. Menon:** When does she respond? When does she ignore? And when she doesn't respond, which time window is responsible? That will tell us when to send our messages.

Nikhil opens the customer engagement dashboard. Chai Theory sends Kavya personalised messages across four time windows every day: morning (6–10 AM), midday (10 AM–2 PM), afternoon (2–6 PM), and evening (6–10 PM). Each message either results in an order or does not.

He realises that the probability of a response is not constant — it depends on when the message is sent. His job is to quantify that pattern and use it to predict future behaviour.

## 1 The Data: Time-Stratified Message Logs

Nikhil pulls up Kavya's message history for the last 60 days. He categorises every message by the time window and whether it resulted in an order.

Table 1: Response rates by time window (last 60 days, total 240 messages)

| Time Window         | Msgs Sent  | Orders Placed | Response Rate |
|---------------------|------------|---------------|---------------|
| Morning (6–10 AM)   | 80         | 28            | 0.35          |
| Midday (10 AM–2 PM) | 60         | 12            | 0.20          |
| Afternoon (2–6 PM)  | 60         | 6             | 0.10          |
| Evening (6–10 PM)   | 40         | 2             | 0.05          |
| <b>Total</b>        | <b>240</b> | <b>48</b>     | <b>0.20</b>   |

Nikhil stares at the table. The pattern is clear: **the later the message, the lower the chance of an order**. During the morning window, Kavya responds 35% of the time. By the

evening, the response rate has dropped to just 5%.

But the overall response rate is only 20%. That means 80% of messages go unanswered. Nikhil wants to understand why.

## 2 Stage 1: The Time-Dependent Response Probability

Nikhil realises that for each time window, the response outcome is a simple yes/no event with a different probability.

### Your Task:

- Identify the probability distribution that models whether a single message results in an order within a given time window.
- For each of the four time windows, write down the probability of an order and the probability of no order.
- For a message sent in the morning (6–10 AM), find the expected value and variance of the response outcome.
- For a message sent in the evening (6–10 PM), find the expected value and variance. Compare the two variances. What does this tell you about the uncertainty of getting a response at different times?

*Round all probabilities to 4 decimal places.*

## 3 Stage 2: A Batch of Morning Messages — Distribution of Orders

Nikhil notices that the brand sends about 10 messages during the morning window over a 10-day period (one per day). He wants to know: if each morning message has an independent 35% chance of resulting in an order, what is the **distribution** of the number of orders?

This is a descriptive question. Nikhil is trying to understand whether a batch of 10 morning messages is likely to produce at least one order, or whether the brand should expect several.

### Your Task:

- Identify the probability distribution that models the number of orders when 10 morning messages are sent.
- Write down the probability mass function.
- Compute  $P(X = k)$  for  $k = 0, 1, \dots, 10$  and tabulate the distribution.
- Compute the expected number of orders and the variance.
- What is the most likely number of orders? What is its probability?
- What is the probability of getting zero orders from the batch of 10 morning messages? Is this unusual?
- Critical thinking:** The binomial model assumes that each message has the same probability  $p = 0.35$  of resulting in an order. Is this assumption realistic? What factors might cause  $p$  to vary from message to message? If  $p$  is not constant, which assumption of the binomial model is violated?

*Round all probabilities to 4 decimal places.*

## 4 Stage 3: The Overall Response Probability

Nikhil now broadens his view. Kavya does not receive messages only in the morning. The brand sends messages across all four time windows. He wants to know: **if a message is sent at a random time, what is the overall probability that it results in an order?**

The law of total probability requires two pieces of information for each time window:

1. **The response probability given the time window**,  $P(\text{order} \mid \text{time}_i)$ . This is the conditional probability that a message results in an order, *given* that it was sent during window  $i$ . We already computed these in Stage 1 from the response rates.
2. **The probability of sending a message during that time window**,  $P(\text{time}_i)$ . This is the proportion of all messages that were sent during window  $i$ . It is *not* a conditional probability — it is simply the relative frequency of messages in that window over the 60-day history.

Nikhil has 240 messages in the dataset. He counts how many were sent in each window:

Table 2: Frequency of messages by time window

| Time Window         | Total Msgs | Proportion     | $P(\text{time}_i)$ |
|---------------------|------------|----------------|--------------------|
| Morning (6–10 AM)   | 80         | 80/240         | 0.3333             |
| Midday (10 AM–2 PM) | 60         | 60/240         | 0.2500             |
| Afternoon (2–6 PM)  | 60         | 60/240         | 0.2500             |
| Evening (6–10 PM)   | 40         | 40/240         | 0.1667             |
| <b>Total</b>        | <b>240</b> | <b>240/240</b> | <b>1.0000</b>      |

**Interpretation:** If Nikhil picks one of the 240 messages at random, there is a 33.33% chance it was sent in the morning, a 25% chance at midday, a 25% chance in the afternoon, and a 16.67% chance in the evening. These proportions are the  $P(\text{time}_i)$  values. Notice that the morning window accounts for one-third of all messages — that is where the brand focuses its engagement.

The two ingredients combine naturally in a tree diagram:

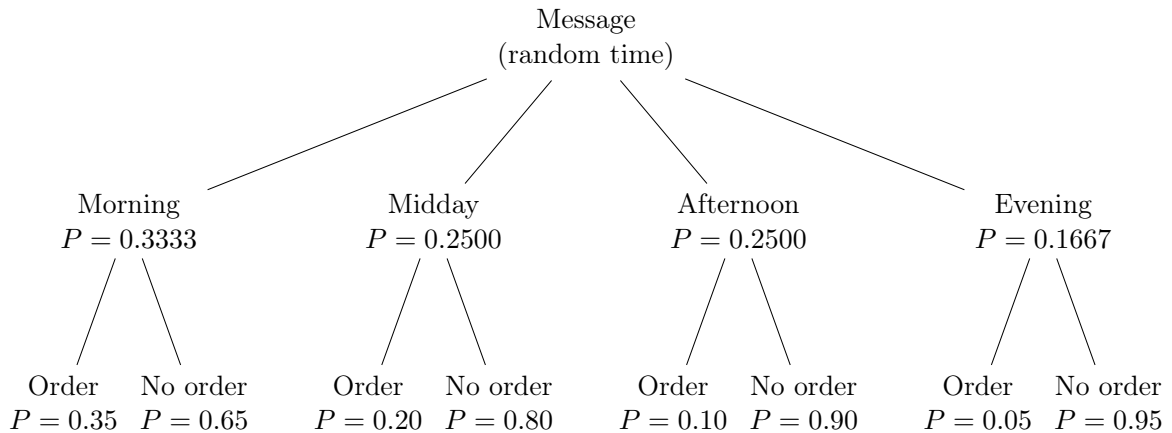


Figure 1: Tree diagram for the law of total probability. The first branch gives  $P(\text{time}_i)$ ; the second branch gives  $P(\text{order} \mid \text{time}_i)$ .

To find the overall probability of an order, we multiply along each branch and add the results:

$$P(\text{order}) = \sum_{i=1}^4 P(\text{order} \mid \text{time}_i) \cdot P(\text{time}_i)$$

**Your Task:**

- Explain in your own words what  $P(\text{time}_i)$  represents. Why is it a proportion, not a conditional probability?
- Verify the values in Table 2. What is the sum of the four proportions?
- Use the law of total probability to compute  $P(\text{order})$ .
- Verify that your answer matches the observed overall response rate of 48/240.
- If the brand sends 20 messages tomorrow at random times (according to the same distribution of times), what is the expected number of orders? What is the variance?
- Critical thinking:** The proportions  $P(\text{time}_i)$  are based on the brand's past 60 days of messaging. Are these proportions likely to remain stable in the future? What might cause them to change?

*Round all probabilities to 4 decimal places.*

## 5 Stage 4: Given No Order, Which Time Window Is Responsible?

Nikhil is now curious about the reverse question: **given that a message resulted in no order, which time window is most likely responsible?**

This is exactly what Bayes' theorem answers. It reverses the conditional probabilities we computed in Stage 3.

**Your Task:**

- Compute  $P(\text{no order})$ , the overall probability of no order. (Hint: Use the complement of the result from Stage 3.)
- Compute  $P(\text{msg sent in evening} \mid \text{no order})$  using Bayes' theorem:

$$P(\text{msg sent in evening} \mid \text{no order}) = \frac{P(\text{no order} \mid \text{evening}) \cdot P(\text{evening})}{P(\text{no order})}$$

- Similarly, compute  $P(\text{afternoon} \mid \text{no order})$ ,  $P(\text{midday} \mid \text{no order})$ , and  $P(\text{morning} \mid \text{no order})$ .
- Compare these posterior probabilities with the prior probabilities  $P(\text{time}_i)$ . What do you observe? Which time window becomes the most likely source of a no-order?
- Based on the posterior probabilities, which time window should the brand prioritise to improve response rates? Explain.
- Using Bayes' theorem, can Nikhil conclude that Kavya is losing interest in the brand? Explain.

*Round all probabilities to 4 decimal places.*

## 6 Stage 5: The Morning Window — Waiting for the First Order

Nikhil realises that the morning window is crucial. If the brand sends a message at 6 AM, it has four hours (until 10 AM) before the response rate drops to the midday level. During this window, multiple messages can be sent, each with a probability of 0.35 of resulting in an order.

Suppose the brand sends 5 messages at regular intervals during the morning window (roughly one per hour).

**Your Task:**

- (a) Identify the probability distribution that models the number of messages until the first order (ignoring the time constraint).
- (b) Find the probability that the first order comes on exactly the 3rd message.
- (c) Find the probability that the first order comes **before 10 AM** — that is, on or before the 5th message.
- (d) Find the probability that no order comes before 10 AM.
- (e) If the brand sends only 3 messages before 10 AM, recompute the probability of getting an order before the window closes. How does the deadline affect the brand's chances?
- (f) What if the brand sends a message at 7 PM (evening window)? What is the probability of an order from a single message? (Hint: Use the evening response rate.)

*Round all probabilities to 4 decimal places.*

## 7 Stage 6: The Rare Event — Evening Orders Over a Month

Nikhil estimates that the brand sends about 30 evening messages to Kavya over the course of a month (one per day). The probability of getting an order from any single evening message is 0.05.

He wants to know: what is the probability that exactly 2 of these 30 evening messages result in orders?

**Your Task:**

- (a) Explain why the Poisson approximation to the binomial is appropriate here.
- (b) Identify the parameter  $\lambda$  for the Poisson approximation.
- (c) Find the probability of exactly 2 orders from 30 evening messages.
- (d) Find the probability of more than 4 orders.
- (e) Based on your calculation, if Kavya placed orders from 5 of the 30 evening messages in a month, would you suspect something is unusual? Discuss.

*Round all probabilities to 4 decimal places.*

## The Realisation

Nikhil leans back in his chair. He has spent the last two hours applying every distribution he knows to a question that has real business consequences.

He has calculated:

- The distribution of orders from a batch of 10 morning messages (Stage 2): zero orders has probability 1.35%.
- The overall response probability across all time windows (Stage 3): 20%.
- The posterior probability that a no-order came from the afternoon window (Stage 4): the highest among all windows.
- The probability of getting an order before 10 AM if 5 morning messages are sent (Stage 5): 88.40%.
- The monthly pattern of evening orders (Stage 6): 5 orders is unusual but not impossible.

The numbers tell him that the morning window is by far the most productive, the evening window is nearly useless, and the overall response rate of 20% hides enormous variation across time windows.

He realises that the data can tell him when a customer is most likely to respond. It cannot tell him why. For that, the brand needs to ask the customer.

He walks to Ms. Menon’s office and presents his findings.

**Nikhil:** The morning window is where Kavya responds. The afternoon and evening windows are almost wasted. We should shift our messaging strategy.

**Ms. Menon:** Then let’s fix that.

## Discussion Questions

1. **Stage 1:** Why is the Bernoulli distribution appropriate for each time window?
2. **Stage 2:** How does the binomial distribution generalise the Bernoulli distribution? What is the probability of getting zero orders from 10 morning messages? Is this unusual?
3. **Stage 3:** What does  $P(\text{time}_i)$  represent, and how is it different from  $P(\text{order} \mid \text{time}_i)$ ? Why is the tree diagram useful?
4. **Stage 4:** In Bayes’ theorem, which time window becomes the most likely source of a no-order after observing the data? Why? Does this match your intuition?
5. **Stage 5:** How does the 10 AM deadline change the geometric distribution? What happens to the probability of an order as the deadline approaches?
6. **Stage 6:** Why does the Poisson approximation work well for evening messages?
7. **Synthesis:** Based on the six stages, what recommendation would you give to Ms. Menon? What assumptions are you making?
8. **Critical Thinking:** Is it ethical to use customer data to predict behaviour? What are the privacy implications? How should the brand use this analysis responsibly?

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*“Data can tell you when a customer is most likely to respond. It cannot tell you why. For that, you need to ask them.”*

— Nikhil, after presenting his analysis to Ms. Menon

## Summary Table (For Students to Fill)

| Stage | Scenario                                        | Distribution | Key Result |
|-------|-------------------------------------------------|--------------|------------|
| 1     | Response probability by time window             |              |            |
| 2     | Distribution of orders from 10 morning messages |              |            |
| 3     | Overall response probability                    |              |            |
| 4     | Given no order, which time window?              |              |            |
| 5     | Order before 10 AM (5 messages)                 |              |            |
| 6     | Evening orders in 30 messages                   |              |            |